Q1 JEE Main 2020 - 8 Jan (Morning)

If c is a point at which Rolle's theorem holds for the function, $f(x) = \log_e\left(\frac{x^2+\alpha}{7x}\right)$ in the interval $[3,\ 4]$, where $\alpha \in R$, then $f''(c)$ is equal to :

(A) $\frac{1}{12}$
(B) $\frac{-1}{12}$
(C) $\frac{1}{6}$
(D) $\frac{-1}{6}$

Q2 JEE Main 2020 - 8 Jan (Morning)

Let the normal at a point P on the curve $y^2 - 3x^2 + y + 10 = 0$ intersect the y-axis at $\left(0,\ \frac{3}{2}\right)$. If m is the slope of the tangent at P to the curve, then $|m|$ is equal to

Q3 JEE Main 2020 - 8 Jan (Morning)

Let $f(x) = x\cos^{-1}(-\sin|x|),\ x \in \left[-\frac{\pi}{2},\ \frac{\pi}{2}\right]$, then which of the following is true?

(A) $f'(0) = -\frac{\pi}{2}$
(B) $f'(x)$ is not defined at $x = 0$
(C) $f'(x)$ in increasing in $\left(\frac{-\pi}{2},\ 0\right)$ and $f'(x)$ in decreasing in $\left(0,\ \frac{\pi}{2}\right)$
(D) $f'(x)$ is decreasing in $\left(\frac{-\pi}{2},\ 0\right)$ and $f'(x)$ is increasing in $\left(0,\ \frac{\pi}{2}\right)$

Q4 JEE Main 2020 - 8 Jan (Evening)

Let S be the set of all function $f : [0, 1] \to R$, which are continuous on $[0, 1]$ and differentiable on $(0, 1)$. Then for every f in S, there exists a $c \in (0, 1)$ depending on f, such that :

(A) $|f(c) - f(1)| < (1 - c)|f'(c)|$
(B) $|f(c) - f(1)| < |f'(c)|$
(C) $\frac{f(1)-f(c)}{1-c} = f'(c)$
(D) None of these

Q5 JEE Main 2020 - 8 Jan (Evening)

Let $f(x)$ be a polynomial of degree 3 such that $f(-1) = 10,\ f(1) = -6,\ f(x)$ has a critical point at $x = -1$ and $f'(x)$ has a critical point at $x = 1$. Then $f(x)$ has a local minima at $x =$

Q6 JEE Main 2020 - 9 Jan (Morning)

A sphere of $10\ cm$ radius has a uniform thickness of ice around it. Ice is melting at rate $50\ cm^3/\min$ when thickness is $5\ cm$ then rate of change of thickness

(A) $\frac{1}{36\pi}$
(B) $\frac{1}{18\pi}$
(C) $\frac{1}{9\pi}$
(D) $\frac{1}{12\pi}$

Q7 JEE Main 2020 - 9 Jan (Evening)

Let a function $f:[0,\ 5]\rightarrow R$ be continuous, $f(1)=3$ and F defined as : $f(x)=\int_1^x t^2 g(t)\,dt$, where $g(t)=\int_1^t f(u)\,du$. Then for the function F, the point $x=1$ is:

(A) a point of local minima
(B) a point of local maxima.
(C) not a critical point
(D) a point of inflection.

Q8 JEE Main 2020 - 2 Sep (Morning)

If $p(x)$ be a polynomial of degree three that has a local maximum value 8 at $x=1$ and a local minimum value 4 at $x=2$; then $p(0)$ is equal to :

(A) 12
(B) 6
(C) -24
(D) -12

Q9 JEE Main 2020 - 2 Sep (Morning)

If the tangent to the curve $y=x+\sin y$ at a point (a,b) is parallel to the line joining $\left(0,\frac{3}{2}\right)$ and $\left(\frac{1}{2},2\right)$, then :

(A) $b=a$
(B) $b=\frac{\pi}{2}+a$
(C) $|a+b|=1$
(D) $|b-a|=1$

Q10 JEE Main 2020 - 2 Sep (Evening)

The equation of the normal to the curve $y=(1+x)^{2y}+\cos^2\left(\sin^{-1} x\right)$ at $x=0$ is :

(A) $y=4x+2$
(B) $y+4x=2$
(C) $x+4y=8$
(D) $2y+x=4$

Q11 JEE Main 2020 - 2 Sep (Evening)

Let $f:(-1,\infty)\longrightarrow R$ be defined by $f(0)=1$ and $f(x)=\frac{1}{x}\log_e(1+x), x\neq 0$. Then the function f :

(A) increases in $(-1,\infty)$
(B) increases in (-1,0) and decreases in $(0,\infty)$
(C) decreases in (-1,0) and increases in $(0,\infty)$
(D) decreases in $(-1,\infty)$

Q12 JEE Main 2020 - 3 Sep (Morning)

The function, $f(x)=(3x-7)x^{2/3}, x\in R$, is increasing for all x lying in :

(A) $(-\infty,0)\cup\left(\frac{14}{15},\infty\right)$
(B) $(-\infty,0)\cup\left(\frac{3}{7},\infty\right)$
(C) $\left(-\infty,-\frac{14}{15}\right)\cup(0,\infty)$
(D) $\left(-\infty,\frac{14}{15}\right)$

Q13 JEE Main 2020 - 3 Sep (Evening)

Suppose $f(x)$ is a polynomial of degree four, having critical points at -1,0,1 . If $T=\{x\in R \mid f(x)=f(0)\}$, then the sum of squares of all the elements of T is :

(A) 4
(B) 2
(C) 6
(D) 8

Q14 JEE Main 2020 - 3 Sep (Evening)

If the surface area of a cube is increasing at a rate of $3.6\mathrm{cm}^2/\mathrm{sec}$, retaining its shape; then the rate of change of its volume $\left(\mathrm{in cm}^3/\mathrm{sec}\right)$, when the length of a side of the cube is $10\mathrm{cm}$, is:

(A) 9
(B) 20
(C) 10
(D) 18

Q15 JEE Main 2020 - 3 Sep (Evening)

If the tangent to the curve, $y = e^x$ at a point (c, e^c) and the normal to the parabola, $y^2 = 4x$ at the point (1,2) intersect at the same point on the x -axis, then the value of c is

Q16 JEE Main 2020 - 4 Sep (Morning)

Let f be a twice differentiable function on $(1, 6)$. If $f(2) = 8, f'(2) = 5, f'(x) \geq 1$ and $f''(x) \geq 4$ for all $x \in (1, 6)$, then :

(A) $f(5) + f'(5) \geq 28$
(B) $f'(5) + f''(5) \leq 20$
(C) $f(5) \leq 10$
(D) $f(5) + f'(5) \leq 26$

Q17 JEE Main 2020 - 5 Sep (Evening)

Which of the following points lies on the tangent to the curve $x^4 e^y + 2\sqrt{y+1} = 3$ at the point (1,0)?

(A) $(2, 2)$
(B) $(-2, 4)$
(C) $(2, 6)$
(D) $(-2, 6)$

Q18 JEE Main 2020 - 5 Sep (Evening)

If $x = 1$ is a critical point of the function $f(x) = \left(3x^2 + ax - 2 - a\right) e^x$, then :

(A) $x = 1$ is a local maxima and $x = -\frac{2}{3}$ is a local minima of f.
(B) $x = 1$ and $x = -\frac{2}{3}$ are local maxima of f
(C) $x = 1$ and $x = -\frac{2}{3}$ are local minima of f
(D) $x = 1$ is a local minima and $x = -\frac{2}{3}$ is a local maxima of f.

Q19 JEE Main 2020 - 6 Sep (Morning)

The position of a moving car at time t is given by $f(t)=at^2+bt+c, t>0$, where a, b and c are real numbers greater than 1. Then the average speed of the car over the time interval $[t_1, t_2]$ is attained at the point:

(A) $2a(t_1+t_2)+b$
(B) $(t_2-t_1)/2$
(C) $a(t_2-t_1)+b$
(D) $(t_1+t_2)/2$

Q20 JEE Main 2020 - 6 Sep (Morning)

Let AD and BC be two vertical poles at A and B respectively on a horizontal ground. If $AD=8m, BC=11\mathrm{m}$ and $AB=10\mathrm{m}$; then the distance (in meters) of a point M on AB from the point A such that $\mathrm{MD}^2+\mathrm{MC}^2$ is minimum is

Q21 JEE Main 2020 - 6 Sep (Evening)

The set of all real values of λ for which the function $f(x)=(1-\cos^2 x)\cdot(\lambda+\sin x), x\in\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ has exactly one maxima and exactly one minima, is :

(A) $\left(-\frac{3}{2}, \frac{3}{2}\right)$
(B) $\left(-\frac{1}{2}, \frac{1}{2}\right)$
(C) $\left(-\frac{1}{2}, \frac{1}{2}\right)-\{0\}$
(D) $\left(-\frac{3}{2}, \frac{3}{2}\right)-\{0\}$

Q22 JEE Main 2020 - 6 Sep (Evening)

For all twice differentiable functions $f: R\rightarrow R$, with $f(0)=f(1)=f'(0)=0$
(A) $f''(0)=0$
(B) $f''(x)=0$, for some $x\in(0,1)$
(C) $f''(x)=0$, at every point $x\in(0,1)$
(D) $f''(x)\neq 0$, at every point $x\in(0,1)$

Q23 JEE Main 2020 - 6 Sep (Evening)

If the tangent to the curve, $y=f(x)=x\log_e x$, $(x>0)$ at a point $(c, f(c))$ is parallel to the line-segment joining the points (1,0) and (e, e), then c is equal to :

(A) $\frac{1}{e-1}$

(B) $\frac{e-1}{e}$
(C) $e^{\left(\frac{1}{1-e}\right)}$
(D) $e^{\left(\frac{1}{e-1}\right)}$

Answer Key

Q1 (A)	**Q2** (4)	**Q3** (D)	**Q4** (D)
Q5 (3)	**Q6** (B)	**Q7** (A)	**Q8** (D)
Q9 (D)	**Q10** (C)	**Q11** (D)	**Q12** (A)
Q13 (A)	**Q14** (A)	**Q15** (4)	**Q16** (A)
Q17 (D)	**Q18** (D)	**Q19** (D)	**Q20** (5)
Q21 (D)	**Q22** (B)	**Q23** (D)	

Q1

$f(3) = f(4) \quad \Rightarrow \quad \alpha = 12$

$f'(x) = \dfrac{x^2 - 12}{x(x^2 + 12)}$

$\therefore \quad f'(c) = 0$

$\therefore \quad c = \sqrt{12}$

$\therefore \quad f''(c) = \dfrac{1}{12}$

Q2

$P \equiv (x_1,\ y_1)$

$2yy' - 6x + y' = 0 \quad \Rightarrow \quad y' = \left(\dfrac{6x_1}{1 + 2y_1}\right)$

$\left(\dfrac{\frac{3}{2} - y_1}{-x_1}\right) = -\left(\dfrac{1 + 2y_1}{6x_1}\right)$

$9 - 6y_1 = 1 + 2y_1 \quad \Rightarrow \quad y_1 = 1$

$\therefore \quad x_1 = \pm 2$

$\therefore \quad \text{Slope of tangent} = \left(\dfrac{\pm 12}{3}\right)$

$= \pm 4$

$\therefore \quad |n| = 4$

Q3

$$f'(x) = x\left(\pi - \cos^{-1}(\sin|x|)\right)$$
$$= x\left(\pi - \left(\frac{\pi}{2} - \sin^{-1}(\sin|x|)\right)\right)$$
$$= x\left(\frac{\pi}{2} + |x|\right)$$
$$f(x) = \begin{cases} x\left(\frac{\pi}{2} + x\right) & x \geq 0 \\ x\left(\frac{\pi}{2} - x\right) & x < 0 \end{cases}$$
$$f'(x) = \begin{cases} \frac{\pi}{2} + 2x & x \geq 0 \\ \frac{\pi}{2} - 2x & x < 0 \end{cases}$$

$f'(x)$ is increasing in $\left(0, \dfrac{\pi}{2}\right)$ and decreasing in $\left(\dfrac{-\pi}{2}, 0\right)$

Q4

If we consider $f(x) =$ constant function, then option (2) becomes $0 < 0$ which is false so answer (2)
cannot be answer. But NTA given by Option (2)

This question was bonus.

Q5

$$f''(x) = \lambda(x - 1)$$
$$\Rightarrow f'(x) = \lambda\frac{(x-1)^2}{2} - 2\lambda$$
$$\therefore\ f(x) = \frac{\lambda(x-1)^3}{6} - 2\lambda x + \mu$$

$$f(1) = -6 \quad \Rightarrow -2\lambda + \mu = -6$$
$$f(-1) = 10 \quad \Rightarrow \frac{2\lambda}{3} + \mu = 10$$
$$\therefore \quad \lambda = 6 \text{ and } \mu = 6$$
$$f'(x) = 0 \quad \Rightarrow x = -1 \text{ or } 3$$
$$\therefore \quad x = 3 \text{ has a local minima}$$

Q6

Let thickness $= x\ cm$

Total volume $v = \dfrac{4}{3}\pi(10+x)^3$

$\dfrac{dv}{dt} = 4\pi(10+x)^2\dfrac{dx}{dt} \quad \ldots\text{(i)}$

Given $\dfrac{dv}{dt} = 50\ cm^3/\min$

At $x = 5\ cm$

$50 = 4\pi(10+5)^2\dfrac{dx}{dt}$

$\dfrac{dx}{dt} = \dfrac{1}{18\pi}cm/\min$

Q7

$F'(x) = x^2 g(x)$

$\Rightarrow \quad F'(1) = 1\cdot g(1) = 0 \quad \ldots(1) \quad (\because \quad g(1) = 0)$

Now $F''(x) = 2xg(x) + x^2 g'(x) \quad (\because \quad g'(x) = f(x))$

$\Rightarrow \quad F''(1) = 0 + 1\times 3$

$\Rightarrow \quad F''(1) = 3 \qquad \ldots(2)$

From (1) and $(2)\ F(x)$ has local minimum at $x = 1$

Q8

Let $p'(x) = \lambda(x-1)(x-2)$

So $p(x) = \lambda\left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x\right] + \mu$

$\because p(1) = 8 \Rightarrow \frac{5}{6}\lambda + \mu = 8$ ...(1)

and $p(2) = 4 \Rightarrow \frac{2}{3}\lambda + \mu = 4$ ...(2)

from (1) and (2); we get $\lambda = 24$ and $\mu = -12$

Now $p(0) = \mu = -12$

Q9

Curve $y = x + \sin y$ and point is (a, b)

$\Rightarrow b = a + \sin b$

Now, $\frac{dy}{dx} = 1 + \cos y\frac{dy}{dx}$

$\Rightarrow \quad (1-\cos y)\frac{dy}{dx} = 1$

$\Rightarrow \left.\frac{dy}{dx}\right](a, b)=\frac{1}{1-\cos b}$ (slope)

Now according to question, $\frac{1}{1-\cos b}=\frac{2-\frac{3}{2}}{\frac{1}{2}-0}$

$\Rightarrow \cos b=0 \Rightarrow b=\frac{\pi}{2}$

Now from (i)

$b=a+\sin b$

$a=b-\sin b=\frac{\pi}{2}-1$

So $|\mathbf{b}-\mathbf{a}|=\left|\frac{\pi}{2}-\frac{\pi}{2}+1\right|=1$

Q10

$y=(1+x)^{2y}+\cos^2\left(\sin^{-1} x\right) \quad \ldots(1)$

On differentiating both sides w.r.t. x we get

$\frac{dy}{dx}=2y(1+x)^{2y-1}+(1+x)^{2y} \ln(1+x)$

$2\frac{dy}{dx}-\frac{\sin\left(2 \sin^{-1} x\right)}{\sqrt{1-x^2}}$

When $x=0$ then $y=(1+0)^{2y}+\cos^2(0)=2$

When $x=0$ and $y=2$ then

$\frac{dy}{dx}=4+0+0=4$

Slope of normal at $x=0$ is $-\frac{1}{4}$

Equation of normal : $y-2=-\frac{1}{4}(x-0)$

$x+4y=8$

Q11

$f'(x)=\frac{\frac{x}{1+x}-\ln(1+x)}{x^2}=\frac{x-(1+x)\ln(1+x)}{(1+x)x^2}<0 \quad \forall x \in(-1, \infty)-\{0\}$

[as $\mathrm{g(x)=x-(1+x)\ln(1+x)}$ gives $\mathrm{g(x)<g(0)}$ for $x \in(-1,0)$ and $g(x)<g(0)$ for $x \in(0, \infty)$]

Q12

$f(x)=(3x-7)\cdot x^{2/3}$

$f'(x)=3x^{2/3}+(3x-7)\cdot\dfrac{2}{3}x^{-1/3}$

$=\dfrac{5x-14}{3x^{1/3}}$

$+\dfrac{x-x+}{0}\dfrac{14}{15}$

$\therefore \quad f'(x)>0$ then $x\in(-\infty,0)\cup\left(\frac{14}{15},\infty\right)$

$\therefore \quad f(x)$ is increasing in $(-\infty,0)\cup\left(\frac{14}{15},\infty\right)$

Q13

Critical points $=-1,0,1$

$\therefore \quad f'(x)=a(x-1)(x+1)x$

$\therefore \quad f(x)=a\left(\frac{x^4}{4}-\frac{x^2}{2}\right)+C$

$\because \quad f(x)=f(0)$

$a\left(\frac{x^4}{4}-\frac{x^2}{2}\right)+C=C$

$a\frac{x^2}{4}\left(x^2-2\right)=0$

$\therefore \quad x=0,\sqrt{2},-\sqrt{2}$

$T=\{0,\sqrt{2},-\sqrt{2}\}$

Sum of square of elements of $T=0^2+(\sqrt{2})^2+(-\sqrt{2})^2$

$=4$

Q14

For cube of side 'a'

$A=6a^2$ and $V=a^3$

Given $\frac{\mathrm{dA}}{\mathrm{dt}}=3.6=12\mathrm{a}\frac{\mathrm{da}}{\mathrm{dt}}$

$\frac{dV}{dt}=3a^2\cdot\frac{da}{dt}=3a^2\left(\frac{3.6}{12a}\right)$

at $a=10$

$\frac{dV}{dt}=9$

Q15

For $y = e^x$
$\frac{dy}{dx} = e^x$
$\left.\frac{dy}{dx}\right|_{x=c} = e^c$
Tangent is $y - e^c = e^c(x - c)$
Put $y = 0, x_1 = c - 1$
For $y^2 = 4x$
$2y\frac{dy}{dx} = 4 \Rightarrow \left.\frac{-dx}{dy}\right|_{y=2} = -1$
Normal is $y - 2 = -1(x - 1)$
Put $y = 0, x_1 = 3$
From (i) and
(ii) ; $\mathbf{c} - 1 = 3$
$\Rightarrow c = 4$

Q16
By using $f'(x) \geq 1$
$\Rightarrow \frac{f(5)-f(2)}{3} \geq 1$
$\Rightarrow f(5) \geq 3 + 8 \Rightarrow f(5) \geq 11$
and also $f''(x) \geq 4$
$\Rightarrow \frac{f'(5)-f'(2)}{5-2} \geq 4$
$\Rightarrow f'(5) \geq 17$
Hence $f(5) + f'(5) \geq 28$

Q17
$x^4 \cdot e^y + 2\sqrt{y+1} = 3$
Differentiating w.r.t. x, we get
$\left(4x^3 + x^4 \cdot y'\right) e^y + \frac{y'}{\sqrt{1+y}} = 0$
$\Rightarrow \quad y'_{\text{at }(1,0)} = -2$
Equation of tangent;
$y - 0 = -2(x - 1) \Rightarrow 2x + y = 2$
Only (-2,6) lies on it

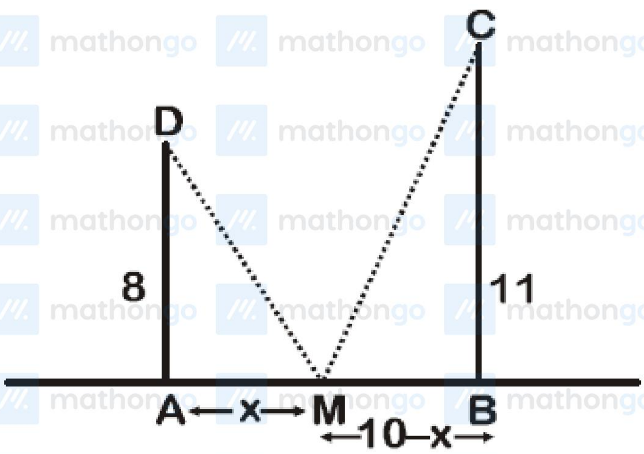
Q18

$f(x)=\left(3x^2+ax-2-a\right)e^x$
$f'(x)=(6x+a)e^x+\left(3x^2+ax-2-a\right)e^x$
$f'(x)=\left(3x^2+(a+6)x-2\right)e^x$
$\because \quad x=1$ is critical point :
$\therefore \quad f'(1)=0$
$(3+a+6-2)\cdot e=0$
$a=-7$
$\therefore \quad f'(x)=\left(3x^2-x-2\right)e^x$
$\qquad\qquad =(3x+2)(x-1)e^x$
$\qquad\qquad +\dfrac{x-y+}{-2/3}+1$
$\therefore \quad x=-\frac{2}{3}$ is point of local maxima.
and $x=1$ is point of local minima.

Q19

Average speed $=\frac{f(t_2)-f(t_1)}{t_2-t_1}=a\left(t_1+t_2\right)+b$
For time at which it is attained
$\frac{df(t)}{dt}=$ Average speed
$2at+b=a\left(t_1+t_2\right)+b$
$t=\frac{t_1+t_2}{2}$

Q20



$\Rightarrow (\mathrm{MD})^2+(\mathrm{MC})^2=64+\mathrm{x}^2+121+(10-\mathrm{x})^2=$

$f(x)$(say)

For minima $\frac{df(x)}{dx}=2x=2(10-x)=0$

$4x=20 \Rightarrow x=5$

$\frac{d^2f(x)}{dx^2}=2-2(-1)>0$

For minimum $x=5$

Q21

$f(x)=\left(1-\cos^2 x\right)(\lambda+\sin x)=\sin^2 x(\lambda+\sin x)$

$\Rightarrow f(x)=\lambda \sin^2 x+\sin^3 x \quad \cdots$ (i)

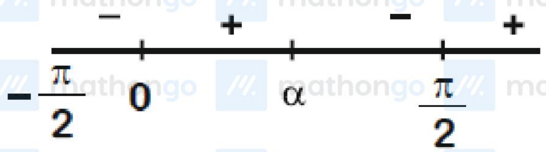
$\Rightarrow f'(x)=\sin x \cos x[2\lambda+3\sin x]=0$

$\Rightarrow x=0$ and $\sin x=-\frac{2\lambda}{3} \Rightarrow x=\alpha$ (let)

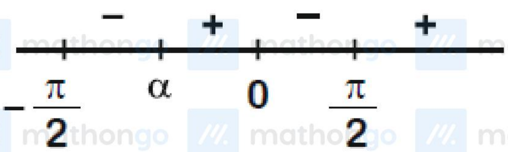
So clearly $f(x)$ will change its sign at $x=0, \alpha$

because there is exactly one maxima and one

minima in $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$



OR



Now $\sin x=-\frac{2\lambda}{3}$

$\Rightarrow \quad -1 \leq -\frac{2\lambda}{3} \leq 1$

$\Rightarrow -\frac{3}{2} \leq \lambda \leq \frac{3}{2}-\{0\}$

$\because \quad$ If $\lambda=0 \Rightarrow f(x)=\sin^3 x($ from (1))

Which is monotonic. so no maxima/minima

So $\lambda \in\left(-\frac{3}{2}, \frac{3}{2}\right)-\{0\}$

Q22

$f: R \rightarrow R$, with $f(0)=f(1)=0$

and $f'(0)=0$

$\because \quad f(x)$ is differentiable and continuous

and $f(0)=f(1)=0$
So by Rolle's theorem
For $c \in(0,1), f'(c)=0$
Now again
$\because \quad f'(c)=0, f'(0)=0$
So by Rolle's theorem
$f''(x)=0$ for some $x \in(0,1)$

Q23

$\because y=x \log_e x \quad(x>0)$
$\Rightarrow \frac{dy}{dx}=1+\log_e x$
$\Rightarrow \left.\frac{dy}{dx}\right]_{x=c}=1+\log_e c \quad$ (slope)
$\because$ The tangent is parallel to line joining $(1,0),(e, e)$
So, $1+\log_e c=\frac{e-0}{e-1}$
$\log_e c=\frac{e}{e-1}-1$
$\Rightarrow \log_e c=\frac{1}{e-1}$
$c=e^{\frac{1}{e-1}}$

mathongo